How machines “Learn” with matrices

- From Neurons to Networks
- Perceptions as logistic regression
- CNNs, RNNs, and Transformer architecture

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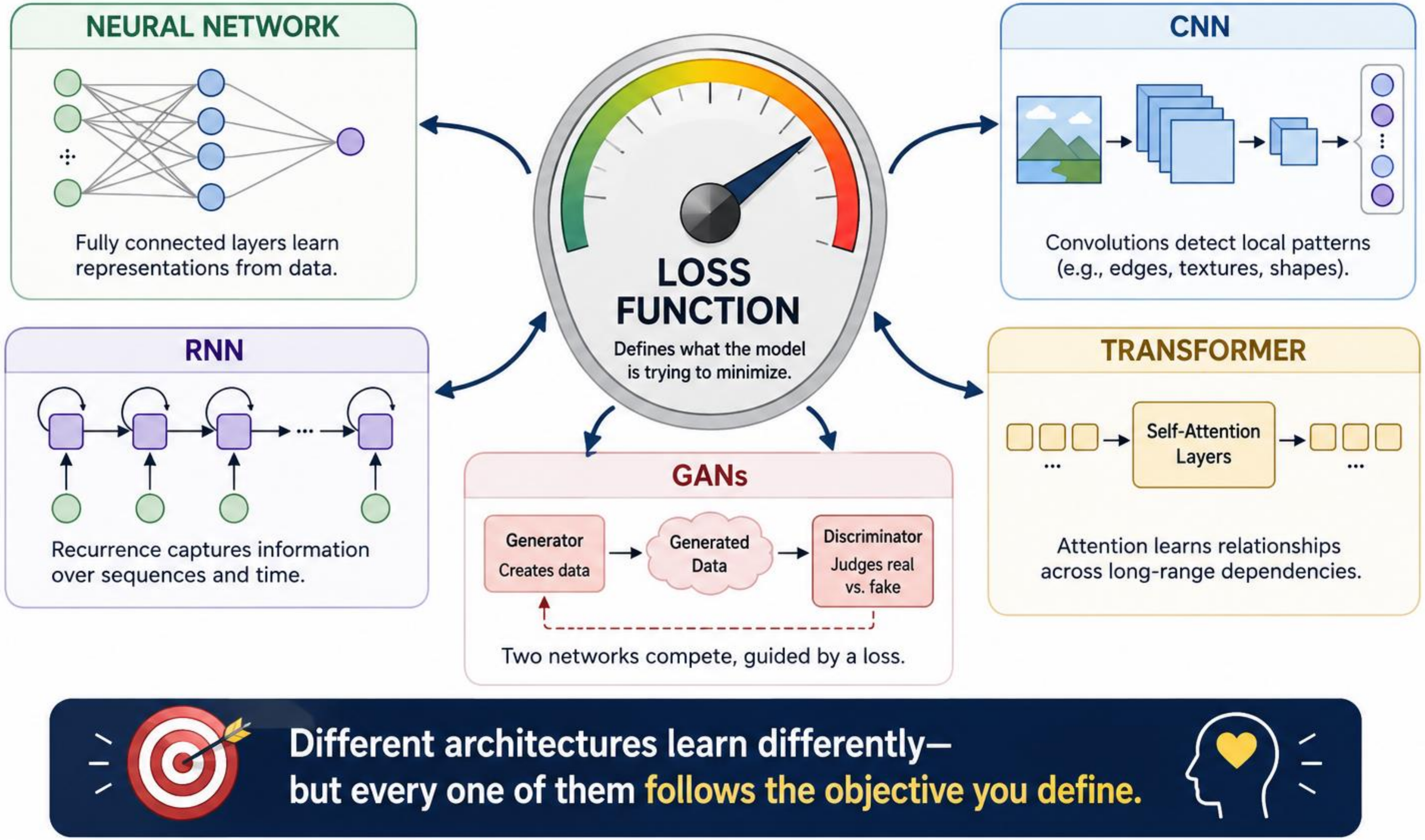
The Loss Function Is Where Alignment Enters

If you measure the wrong thing, the model optimizes towards the wrong goal!



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To discuss after the activity

This activity demonstrates how a fully connected network operates

There was one part of this exercise will look ahead to our next topic. Can anyone identify it?

Can you think of any ways to improve our HNN's performance?

Human Neural Network Activity

No speaking during the rounds

1. We will assign each of you to one of the following roles:

	Group A	Group B
Input Layer (4 people)		
Hidden Layer (5 people)		
Output Layer (1 person)		

2. **Input layer people** will go to breakout Zoom room and see an image
3. **Input layer people** will write down **FIVE** words that describe the image in their assigned table in the google document.
4. **Hidden layer people** will choose **TWO** words of the **FIVE** words that best summarize the image
5. **Output layer person** will use the **TWO** words from each hidden layer of their group and create a short summary of the image

Human Neural Network Activity

No speaking during the rounds

- We will assign each of you to one of the following roles:

	Group A	Group B
Input Layer (4 people)		
Hidden Layer (5 people)		
Output Layer (1 person)		

- **Input layer people** will write down **FIVE** words that describe the image in their assigned table ROW in the google document.
- **Hidden layer people** will choose **TWO** words of the **FIVE** words that best summarize the image and put these words in the SYNTHESIZE PAGE
- **Output layer person ONLY** uses the **two-word** summaries on synthesis page to create their description

Activity Discussion

This activity demonstrates how a fully connected network operates

There was one part of this exorcise will look ahead to our next topic. Can anyone identify it?

Can you think of any ways to improve our HNN's performance?

Learning Goals:

What are the different types of AI model and how are they used

You will be able to:

- Break down a model into its basic layers
- Identify what type of layer sets each model apart
- Understand the purpose of each type of layer

Neural Networks: Perceptrons & Fully connected Networks

Not enough regression

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Perceptrons & Fully connected Networks

Learning Goals:

What are neural networks and how do they work?

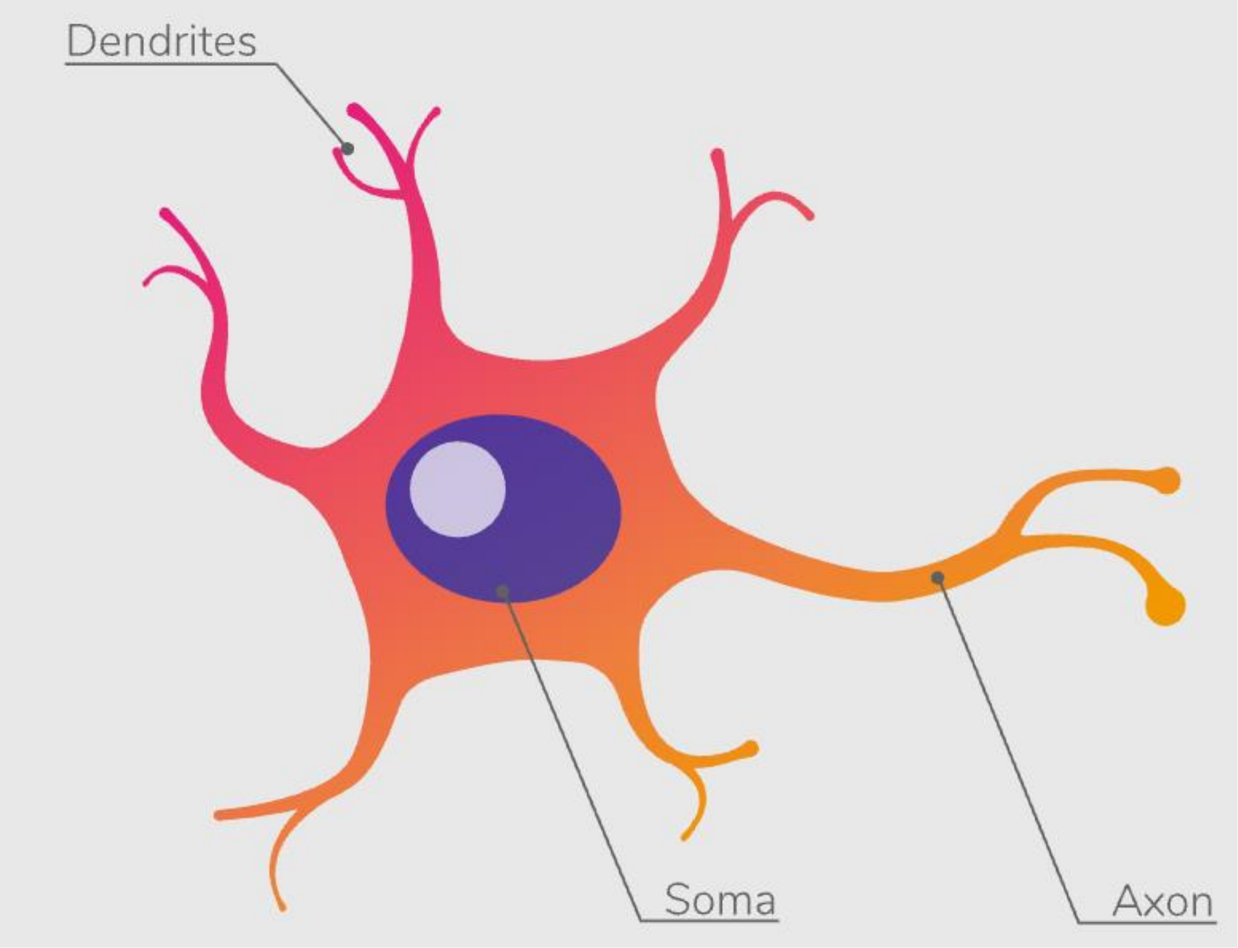
You will be able to:

- Explain what a perceptron is
- Combine perceptrons into a fully connected layer

A Brief Aside: Biological Neurons

Neurons are the foundational unit of the nervous system.

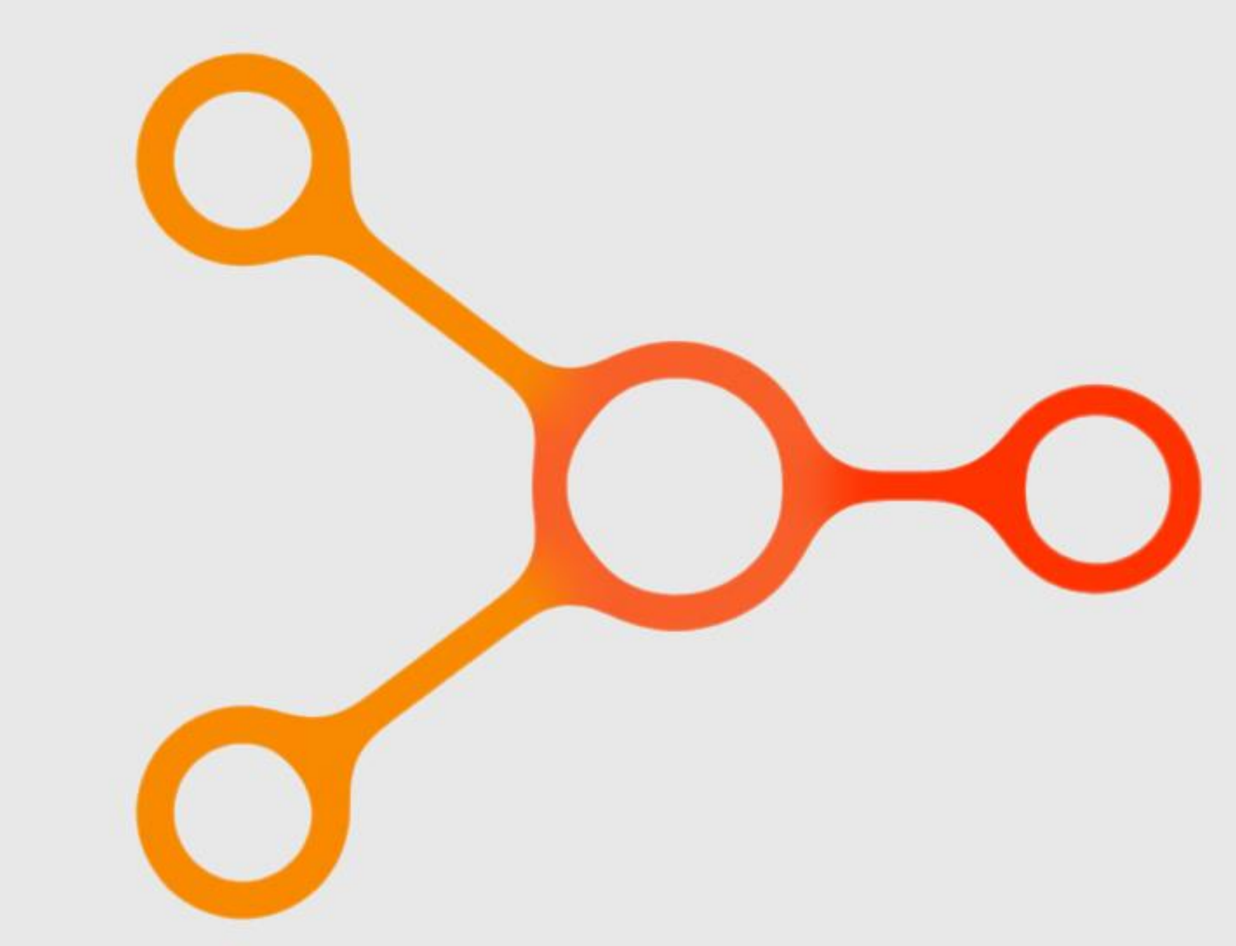
- Dendrites take in signals from other cells
- The cell body/soma processes these signals and decides whether to “fire”
- When the neuron fires a signal is sent through the axon to other cells



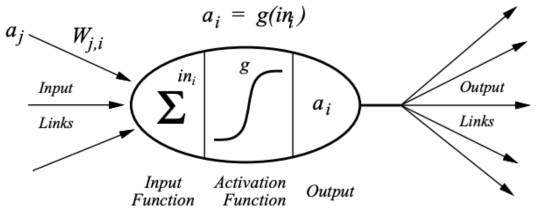
Can we create a mathematical model based on neurons?

Perceptrons are the foundational unit of neural networks.

- A perceptron takes one, or (usually) more, inputs.
- These inputs are combined, using a weighted sum, into a single output.
- This output can then be modified by an activation function
- You may have noticed the behavior of an individual perceptron is basically a regression model like we saw yesterday

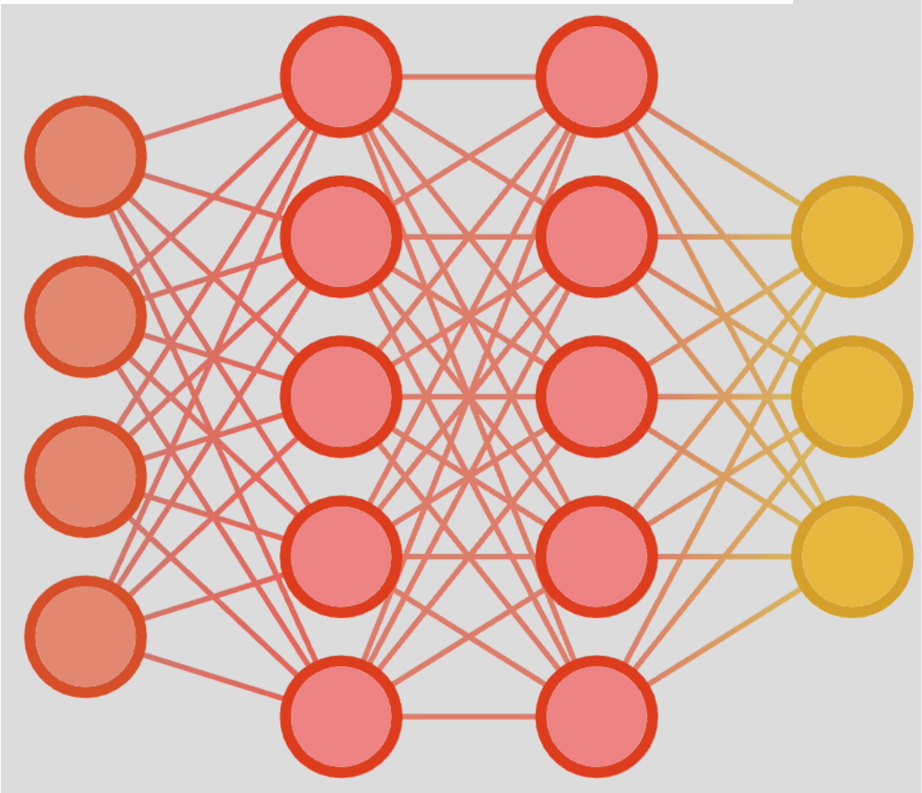


We’ve recreated regression, so what?



Age	Glucose	Has Diabetes (Yes = 1, No = 0)
40	130	1
25	100	0

$$a_i = g(\sum_j W_{j,i} a_j)$$



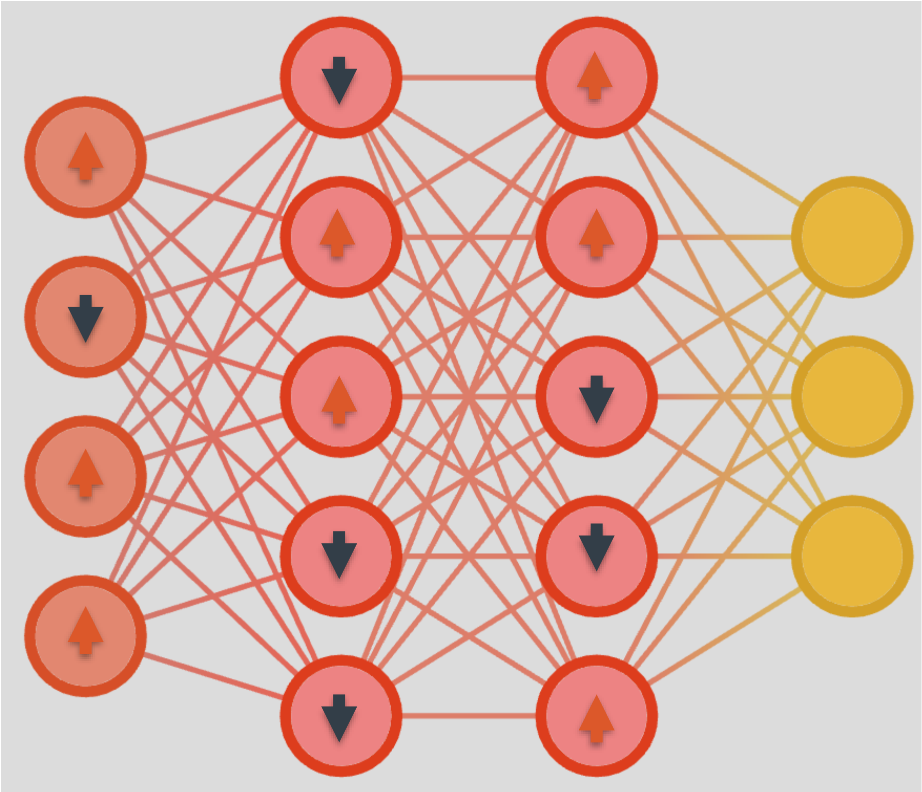
The brain is not made up of just one neuron

- By connecting multiple perceptrons together we create an artificial neural network
  - Two important terms: Fully connected & Feed forward
- Artificial neural networks can solve much more complex problems than “simple” regression

How do we learn the weights?

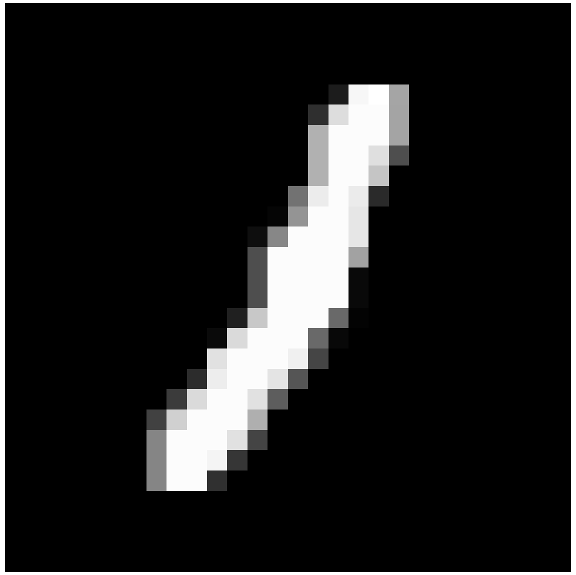
Remember gradient decent?

- We now have a much more complex function
- The chain rule from calculus let’s us break up a complex derivative
- We can see how each perceptron is contributing to the gradient and determine how to address its weights
- This is called “Back propagation”



What are some major limitations of Fully Connected Layers?

1. Scalability
2. Scalability!
3. Have I mentioned scalability?
4. Restricted Input



vs

Many networks end in one or more fully connected layer(s).

What we discuss the rest of today and tomorrow will by and large be ways to take input that is too large/amorphous to go directly into a Fully connected network and finding a "feature representation" that accurately represents that input

Review

Fully connected Networks

- Built from multiple regression models combined together
- Can solve complex non-linear problems
- Still used as a fundamental part of many more advanced types of network
- Need to represent input data as a vector of fixed size

10 Minute break

Returning at x:xx

Next up: Convolutional Neural Networks

Neural Networks: Convolutional Neural Networks

Finding patterns in matrixes

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Learning Goals:

What are Convolutional Neural Networks (CNNs) and how do they build on fully connected neural networks ?

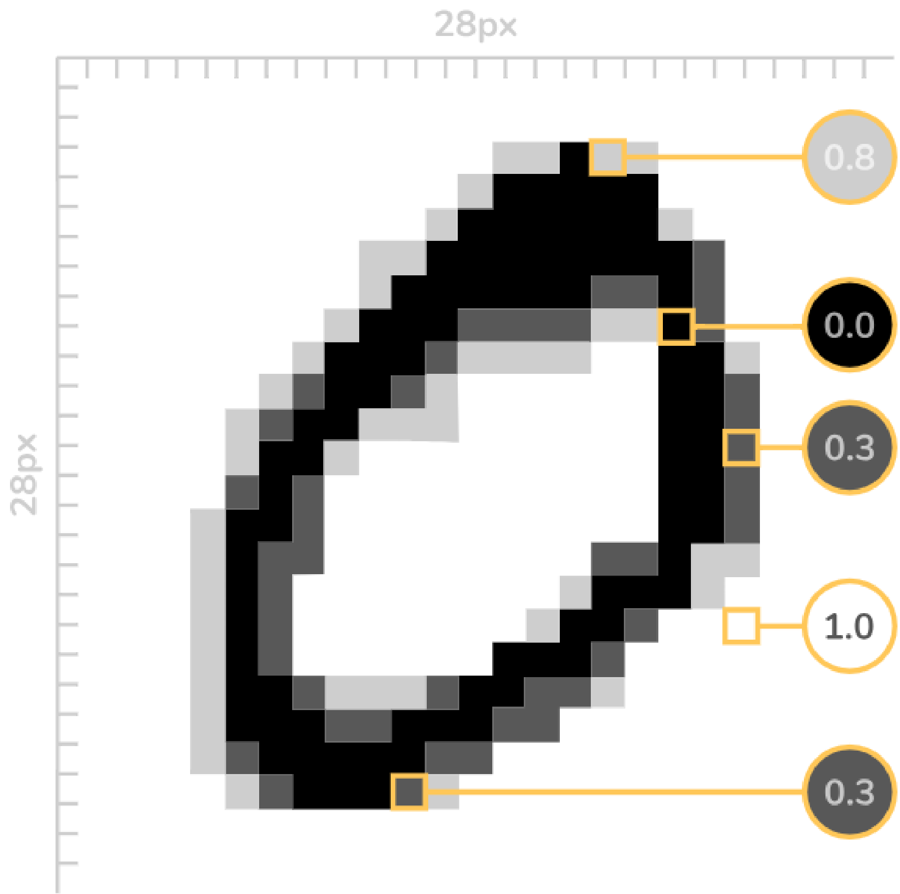
You will be able to:

- Explain what a convolution does
- Contrast a convolutional layer to a fully connected layer
- Determine when to use convolutional layers

A brief aside: An image is worth HxW numbers

Think of your computer monitor

- Anything you see on it is thousands of little points of light (pixels)
- Color is just a combination of red green and blue points of light
- We know how bright each point is
- We can represent any image as a rectangle of numbers aka a matrix



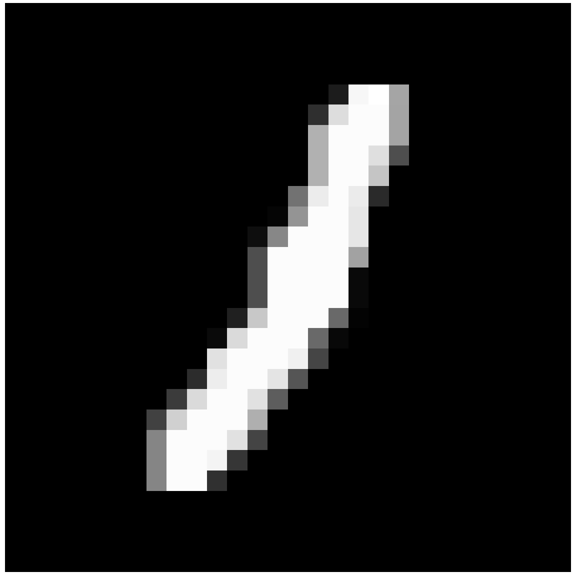
Why do we need convolutions?

MNist has 28pxl X 28 pxl gray scale images

- Each image can be flattened into a 784-element vector

This picture of a dog is 512pxls X 910pxls and in color.

- If we flattened it, it would be nearly five hundred thousand numbers
- We need a way to accurately describe the image with far fewer numbers



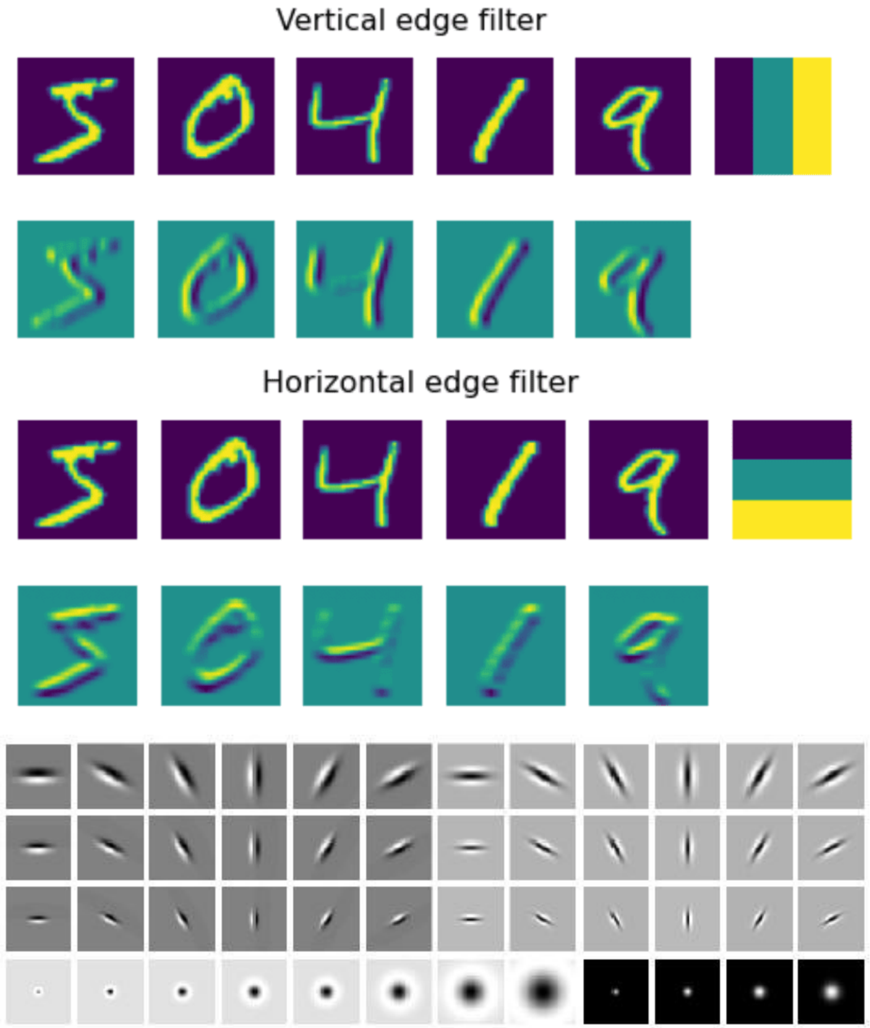
vs

What is a convolution?

Convolutions are filters that look at small patches of an image to find patterns

- We can create filters manually to find specific shapes
    - This is what the input layer was doing in the HNN activity

- Instead of telling the network what to look for, we can let it learn what shapes are important just like we did with the perceptrons



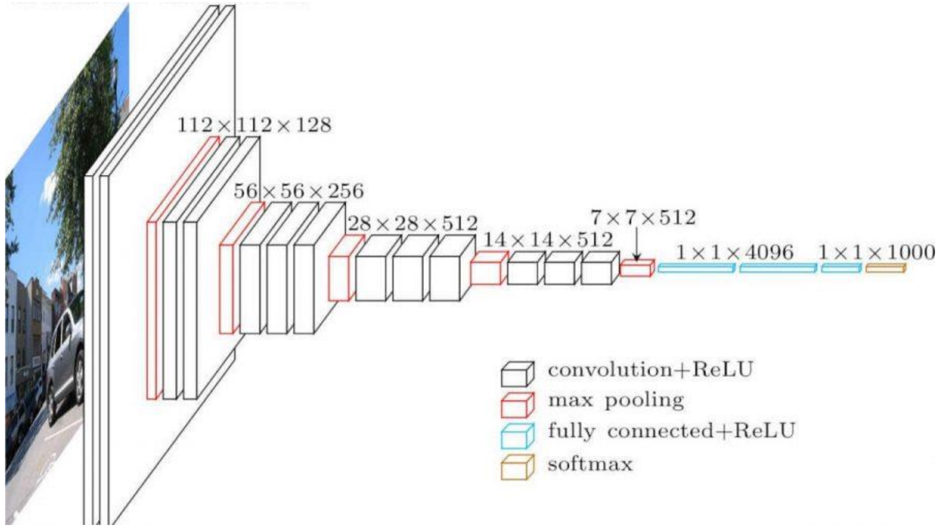
How do we find more complex shapes?

Multiple perceptrons ➔ Fully connected layer

Multiple FCs ➔ Fully connected network

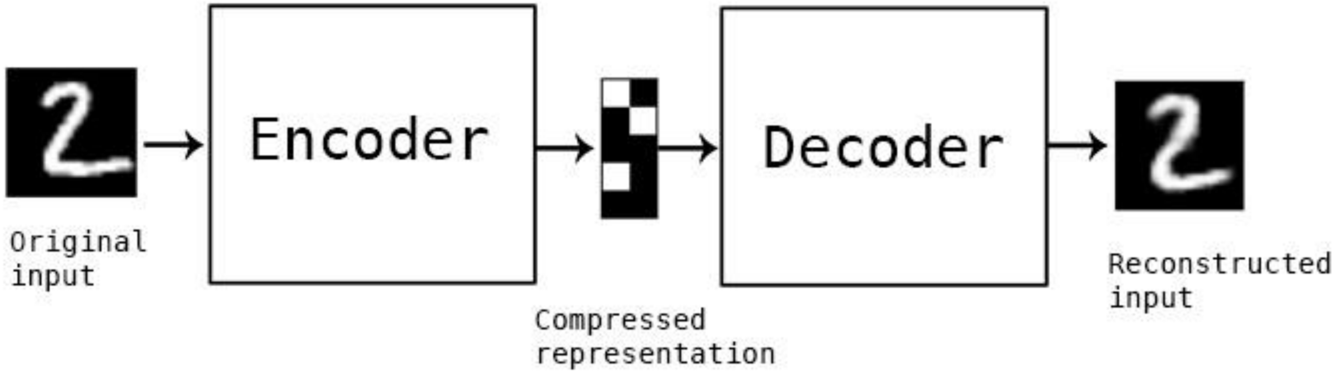
Multiple convolutions ➔ Conv layer

Multiple conv layers ➔ Convolutional NN



What about going in reverse?

Deconvolutions are the inverse of convolutions



- Convolutions take a series of patterns and turn an image into a description of that image

- Deconvolutions take a description and “draw” an image based on it.

- An alternative to ending a network in a fully connected layer when you want a matrix instead of a vector. E.g. image generation

To discuss after the activity

This game demonstrates using simplified patterns to describe complex images

What kinds of shapes were more informative?

Were shapes from certain parts of the image more informative than others?

Contour Classification Activity

1. Go to the following PowerPoint:
    - AI Literacy 2026 Contour Game.pptx

2. These are a series of puzzles based off traces from pictures of animals.
3. The even numbered slides add more contours/shapes which may be more informative
4. Feel free to move the shapes around, they have not been rotated
5. Can you guess which animal was used to create each set of shapes

Activity Discussion

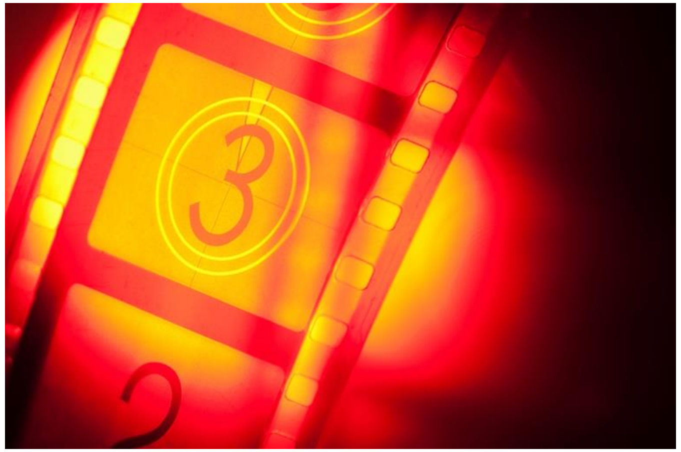
This game demonstrates using simplified patterns to describe complex images

What kinds of shapes were more informative?

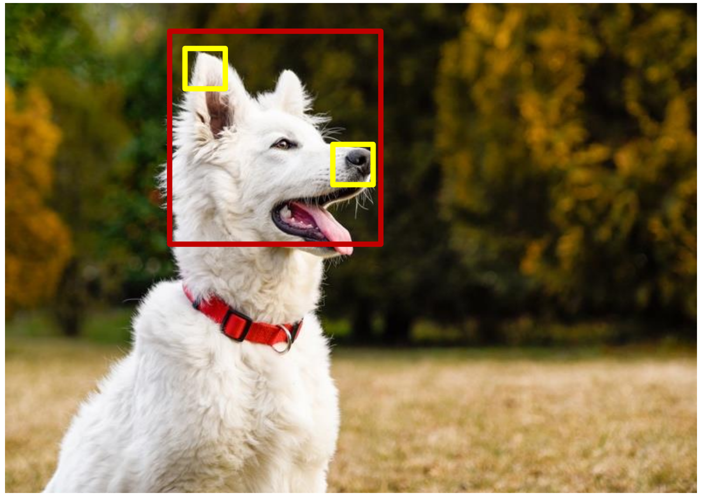
Were shapes from certain parts of the image more informative than others?

What are some major limitations of convolutional layers?

1. No memory
   - Imagine working with a video instead of an image
   - There is temporal context the convolution can not see



2. Limited field of view
   - Because of the limited size of a convolution, you have limited context for fine details



3. Fixed number of channels

Review

Images can be thought of as matrixes of numbers

Convolutional neural networks

Strengths:

- Can efficiently summarize large input matrixes
- Can be reversed to (re)create an image

Limitations:

- No memory
- Limited field of view
- Fixed number of channels

10 Minute break

Returning at x:xx

Next up: Recurrent Neural Networks & Natural language Processing

Neural Networks: Recurrent Neural Networks

Working with Sequences

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Learning Goals:

What are Recurrent Neural Networks (RNNs) and how do they allow us to work with sequential data such as text?

You will be able to:

- Tokenize input data
- Explain why you would add a loop to a network
- Infer the limitations/downsides of Recurrent layers

A brief aside: Tokenization

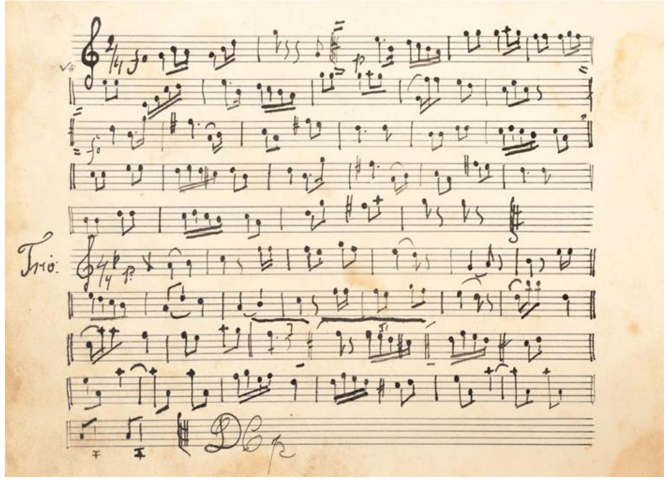
Q: How do we perform calculations on data that isn't numbers?

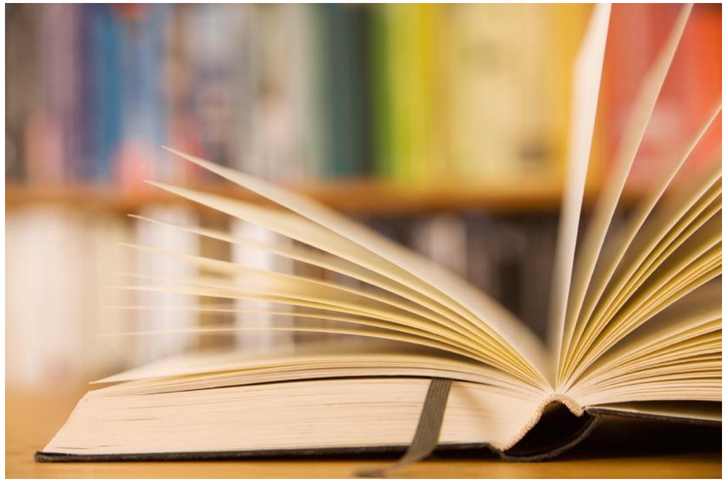
A: A bit of a trick question, because we can't do so directly.
But we can use numbers to represent non-numeric data.

Types of tokenization used in AI/ML:

- Word level
- Character level
- Sub-word level

What sorts of data are sequential?





How is Sequential data different?

Order Matters

- Earlier parts of sequential data provide context for later parts
- Imagine hearing the punchline of a joke before the setup (or without the set up at all)
- Consider how the meaning of this sentence changes if we change the order of two words

Eve
spied
on
Alice

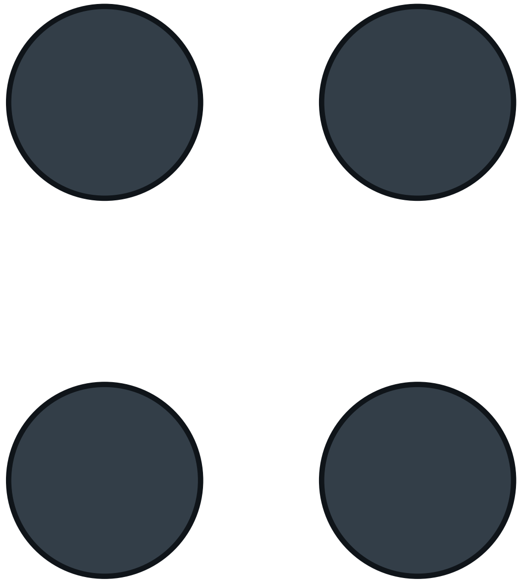
What problems does this create?

We often don’t know the size of our input

- Two sentences can have different numbers of words.
- This means each token must be considered individually

“feed forward” networks only see a snapshot of what is going on

- They possess no memory of previous inputs

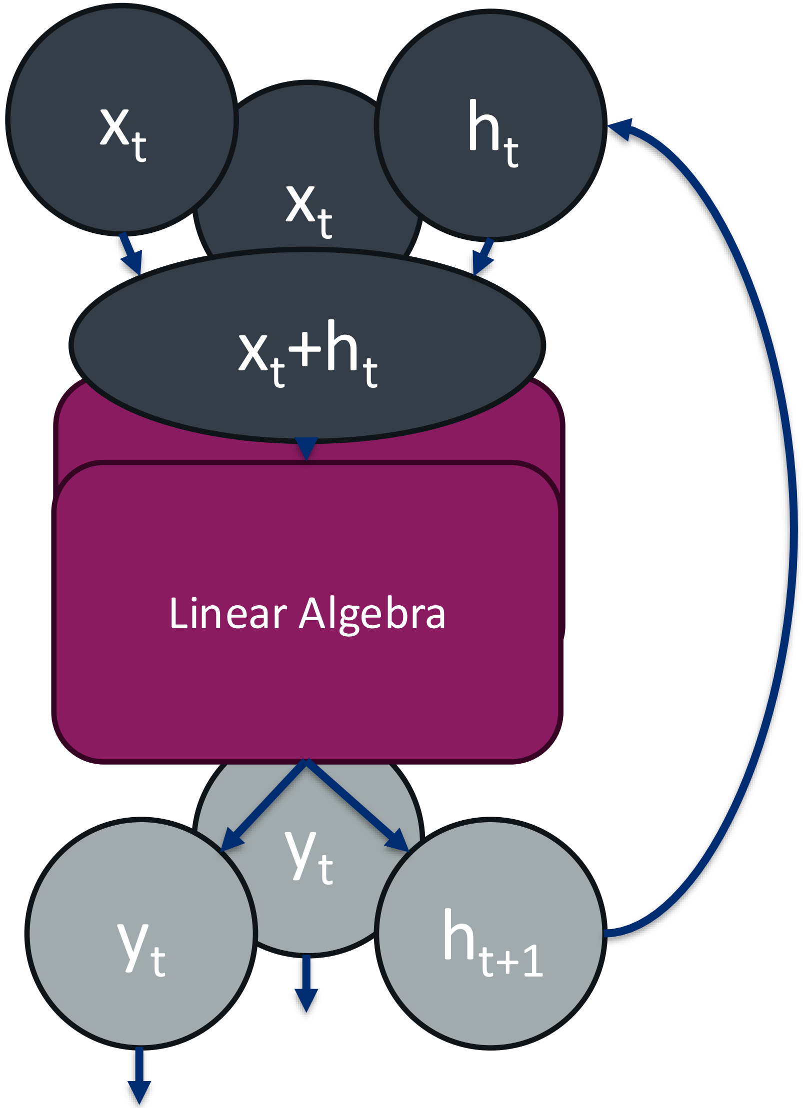


How can we give our network a memory?

Recurrent layers to the rescue

We create a new vector **h** that can store information for the layer

Each input **x** is connected to **h** and fed through the layer to create an output **y** and a new **h** that is used with the next input



To Discuss after the activity

This game demonstrates how considering the sentence as a sequence improves interpretability

Can you think of any limitations with considering the sentence in this way?

Can you think of a method that would improve on this sequential processing?

Sentence Analysis Activity

1. Break into small groups
2. Go to the following link:
    - AI_Literacy_Sentence_Game_Final.pptx
3. Collapse the slide preview on the left
4. These are a series of sentences proceeded by a random word from the sentence. Can you guess the meaning of the sentence from the random word?
5. Go through the sentence word by word, how many words does it take you to figure out the meaning?

Activity Discussion

This game demonstrates how considering the sentence as a sequence improves interpretability

Can you think of any limitations with considering the sentence in this way?

Can you think of a method that would improve on this sequential processing?

What are some major limitations of recurrent layers?

1. Limited memory
   - There is an upper limit to how much information can be stored in **h**
   - This puts an upper limit on number of tokens the network can consider
2. Recency bias
   - Because **h** changes with each input the networks memory prioritizes more recent tokens
3. No token prioritization
   - The network has no way of learning which tokens are more important
4. Limited parallelizability
   - Whenever you treat something as a sequence you have to process it in order limiting your ability to split the work across multiple processors

What are some major limitations of recurrent layers?

1. Limited memory
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VS

What are some major limitations of recurrent layers?

2. Recency bias
   - Because **h** changes with each input the networks memory prioritizes more recent tokens

The quick brown fox jumps over the lazy dog

What are some major limitations of recurrent layers?

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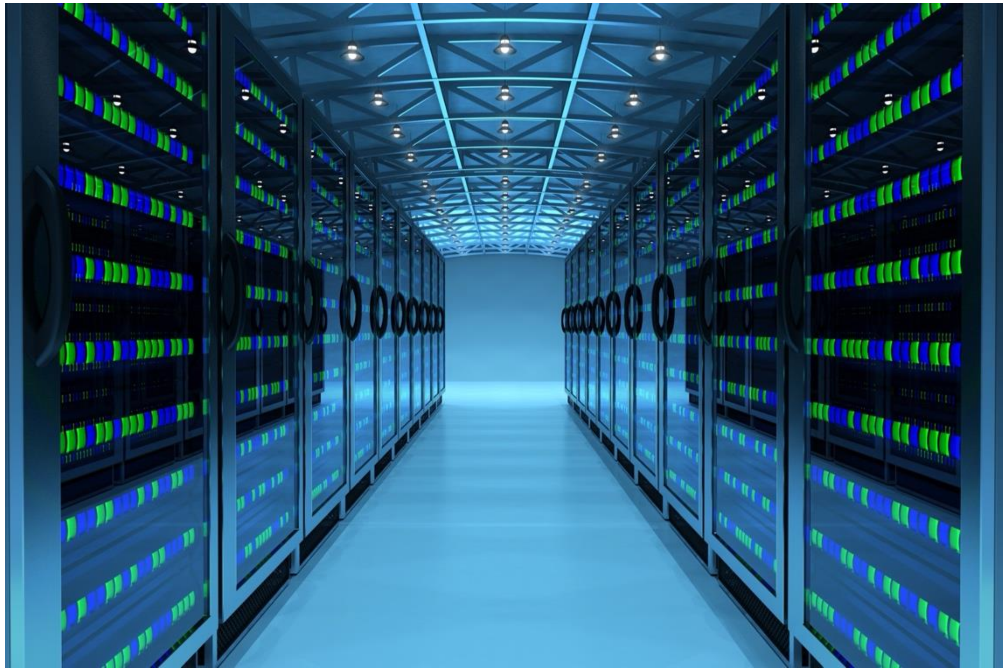

Review

Tokenization

- By representing things like words with vectors of numbers we can input them into neural networks

Recurrent neural networks

Strengths:

- Adds memory to a neural network
- Incorporates sequential context
- A lot of things are sequential

Limitations:

- Computationally expensive
- Limited memory capacity and range
- Can’t learn relative importance of tokens

Neural Networks:
Generative Adversarial Networks (GANs)

Turning Generation into prediction

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Learning Goals:

How do we Generate new things with a neural network and what does a state-of-the-art network look like?

You will be able to:

- Contrast generative vs. discriminative / predictive models; outline GAN generator–discriminator interplay.
- Explain at a high level how self-attention works and why it replaced recurrence for many tasks.
- Appreciate the power (and pitfalls) of large-scale pre-training through guest talk & games.

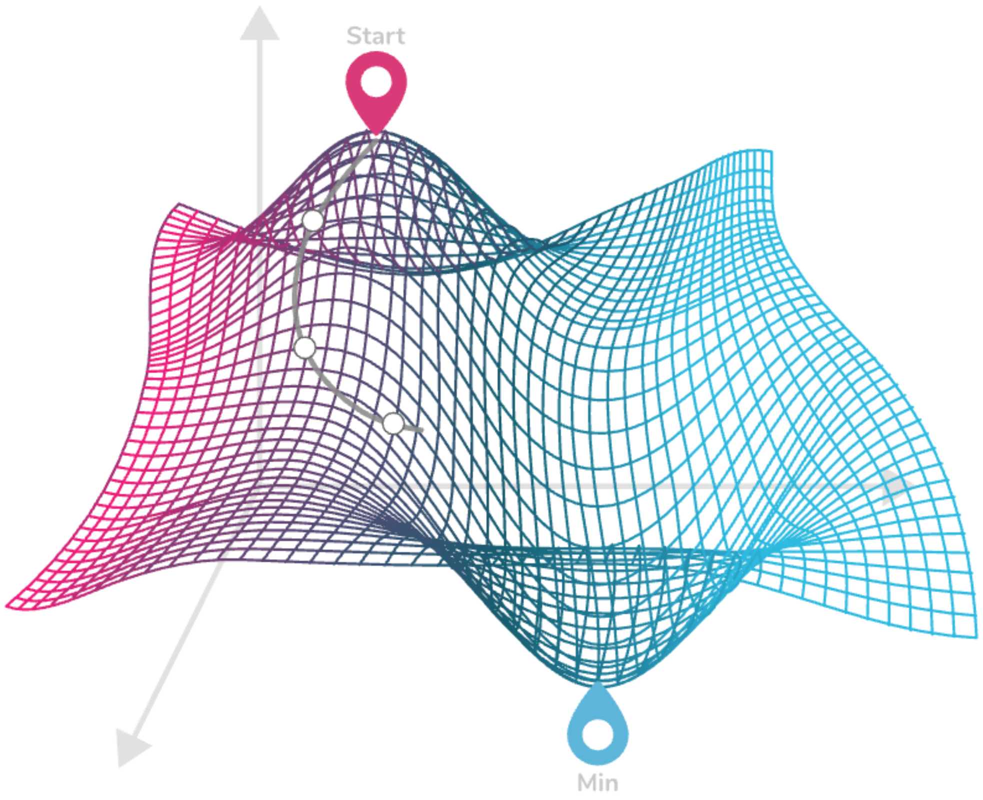
Call Back: Loss Functions

A good loss function is necessary to train a neural network

- Think back to the treasure splitting game from day 1

- A loss function measures how different your output is from the desired output

- This is easy when you have a single right answer for each input



How do we quantify the quality of generated content?

Q: When generating something "new", how can we compute the loss?

- What are we trying to quantify?
  - Plausibility

- How can we measure that?
  - What if we could measure how good our generator is at tricking someone or something?

To consider: what implications does defining loss in this way have for AI Hallucinations

Can we turn generation into a prediction problem?

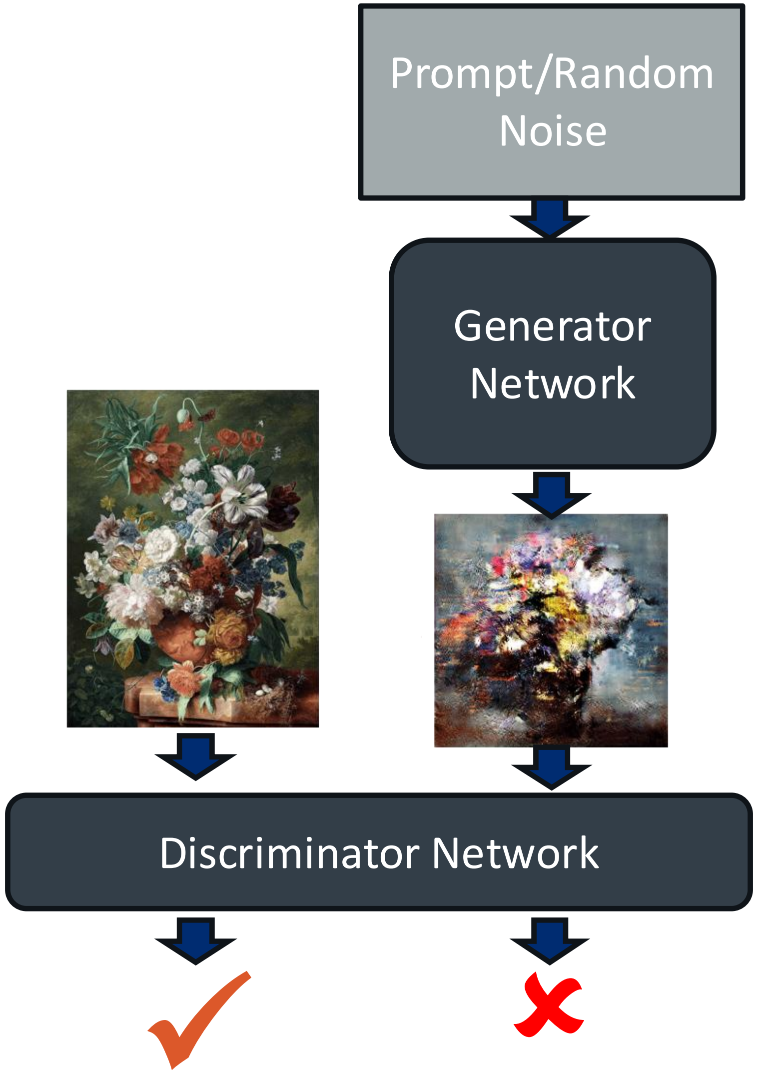
Generative Adversarial Networks have two parts:

- The Generator which creates something new

  Examples:

  - Deconvolutions to create an image
  - A Recurrent Network that chooses a plausible next word for a sentence

- The Discriminator which takes the generated output and a real example and tries to determine which is the fake



To discuss after the activity

This activity puts you in the role of generators and discriminators

Generators: what strategies did you use to improve the plausibility of the answers as the game went on?

Discriminators: were you basing your decision on anything other than plausibility?

Everyone: what implications would training a network in this way have, if there were no correct answer?

Data science balderdash

1. Split into pairs and decide on who will be the generator and who will be the discriminator. (in the case of an odd number of players one group should have 2 discriminators)
2. Go to the following PowerPoint, and collapse the slide preview on the left:
    - AI_Litteracy_2026_DS_Balderdash.pptx
3. The slides are split into pairs: a data science term, followed by its definition
4. Generators:
    - Make up a definition for the term
    - Tell the discriminator(s) both your fake definition and the real definition
5. Discriminators:
    - Guess which definition is the real one
6. Keep playing more rounds until time is called or you run out of slides (optionally switching roles)

Activity Discussion

This activity puts you in the role of generators and discriminators

Generators: what strategies did you use to improve the plausibility of the answers as the game went on?

Discriminators: were you basing your decision on anything other than plausibility?

Everyone: what implications would training a network in this way have, if there were no correct answer?

What are some major limitations of GANs?

1. Limited by the capabilities of the generator and discriminator architectures
2. Hard to train and can't be "transferred" easily

Only a limitation in some cases:

3. Correctness is only rewarded in so far as it makes the generated answer more plausible
4. Will always generate an output

vs

Review

Generative Adversarial Networks

- Allows quantifying loss for generated output
- Turns a generation problem into a classification problem
- Trained to produce plausible output
- Will always produce some form of output

Neural Networks: Transformers & Self-Attention

Adding context

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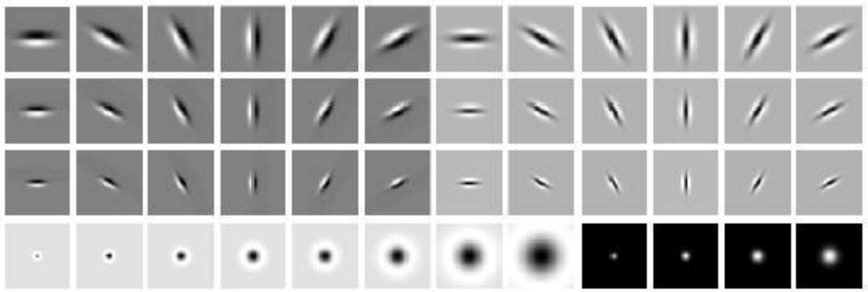
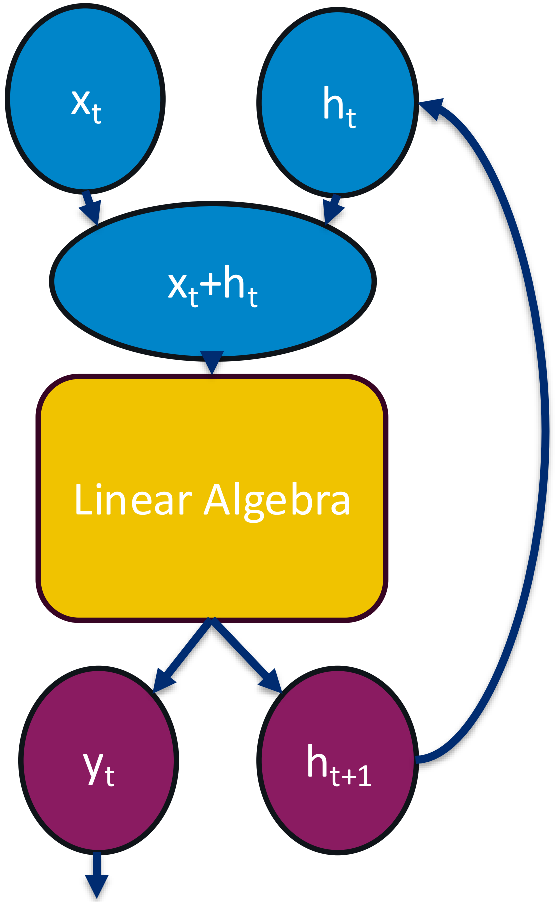
Context is king



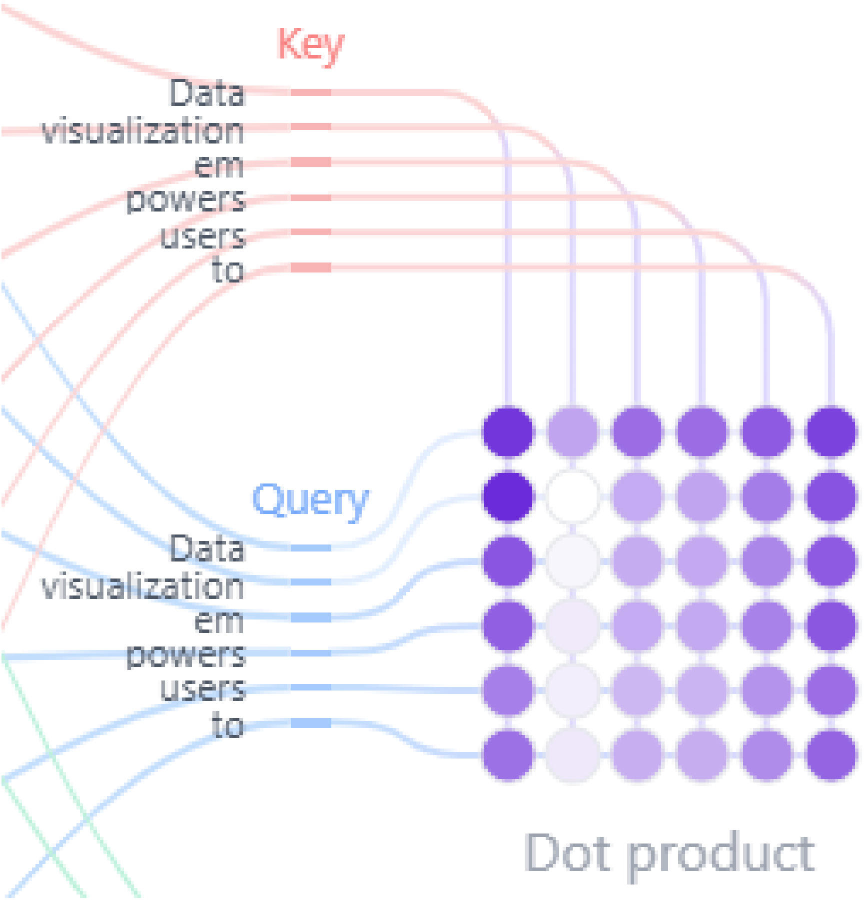
- What do convolutions and recurrent layers have in common?
- Capture context more efficiently than fully connected layers.
- Convolutions summarize data by finding patterns.
- Recurrence summarizes sequences into a single data point via cumulative memory.
- Both focus on local context



How do we broaden our context window?

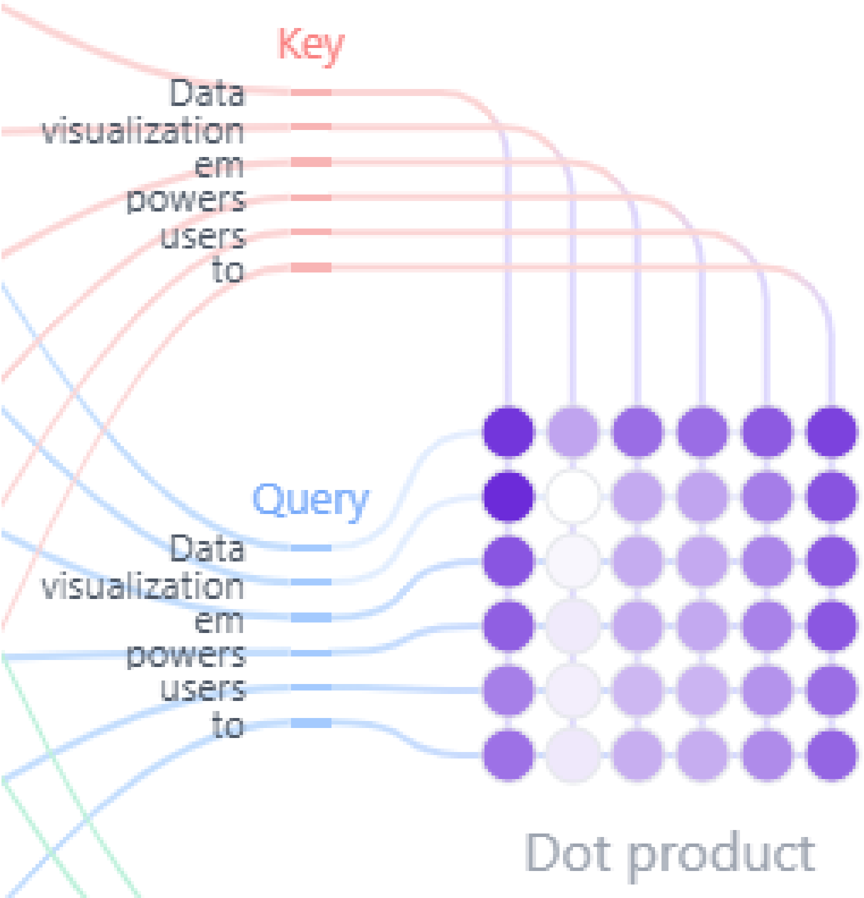
Self attention captures the relationships between input tokens and has 3 steps:

1. Measure the similarity between tokens
2. Scale the similarity using learned weights
3. Apply an activation function & regularize

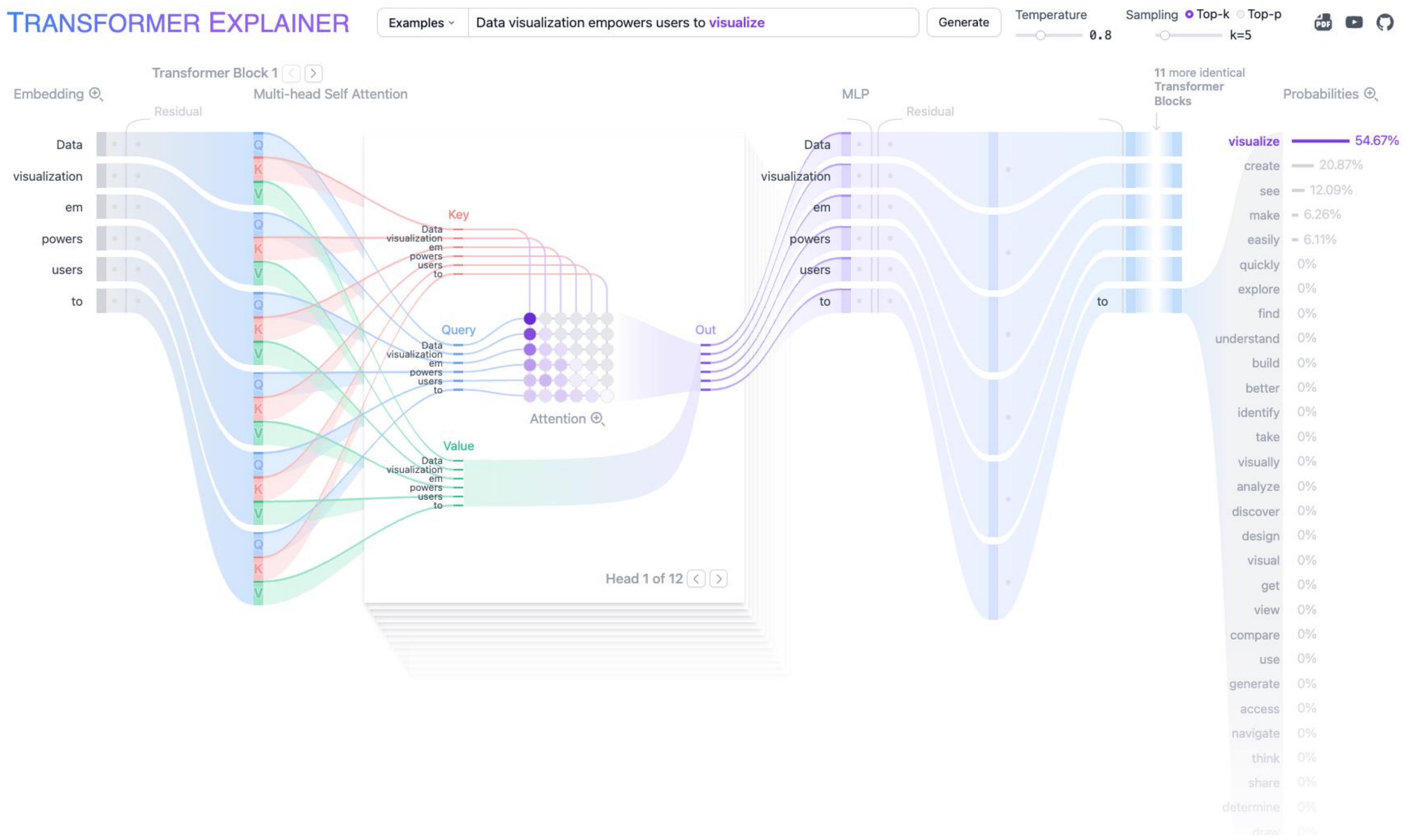


What are the advantages of self-attention?

- Can connect separated words if they are contextually related
- Multiple “attention heads” can capture different relationships
- Each attention head can be calculated in parallel



How do transformer work in practice?

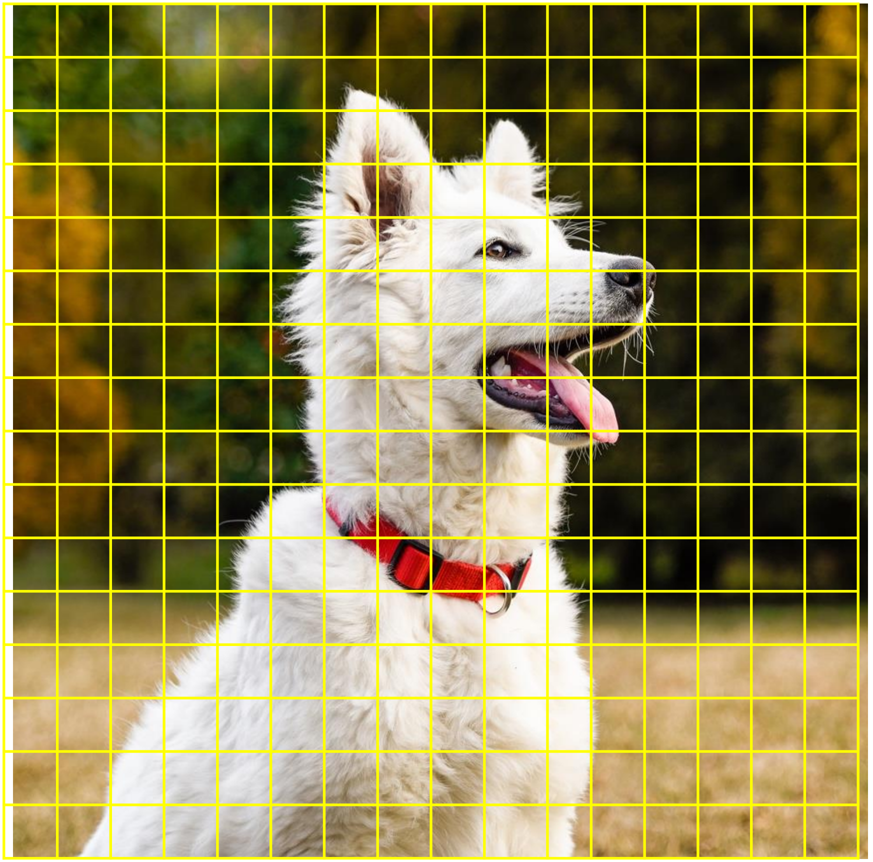


https://poloclub.github.io/transformer-explainer/

Brief aside: Tokens aren't just for words

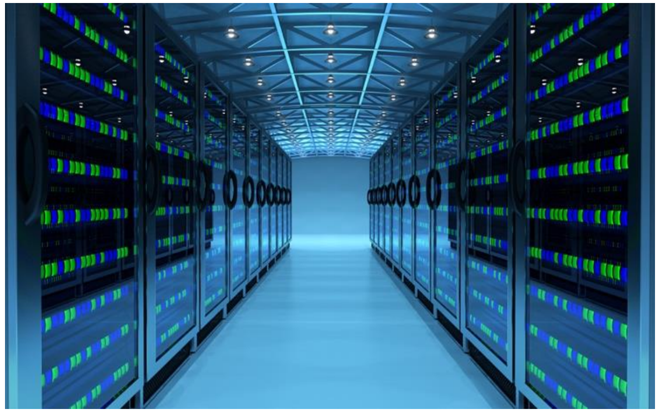
Vision transformers use the same self attention mechanism to improve image analysis

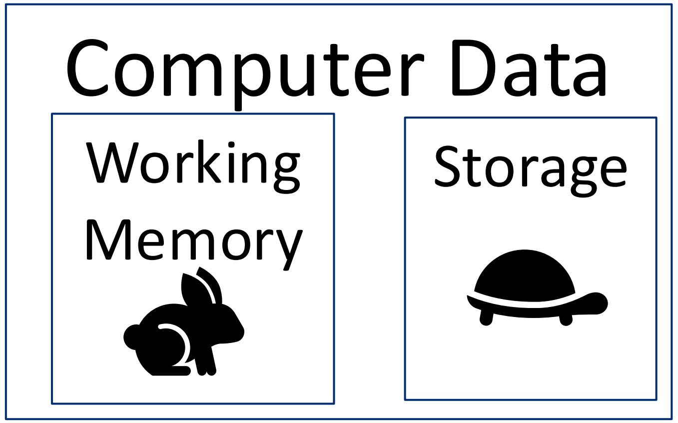
- Much like tokenizing a sentence an image is broken down into patches
- Convolutions are used to turn each patch into a feature vector
- The feature vectors are used as tokens by the attention layer

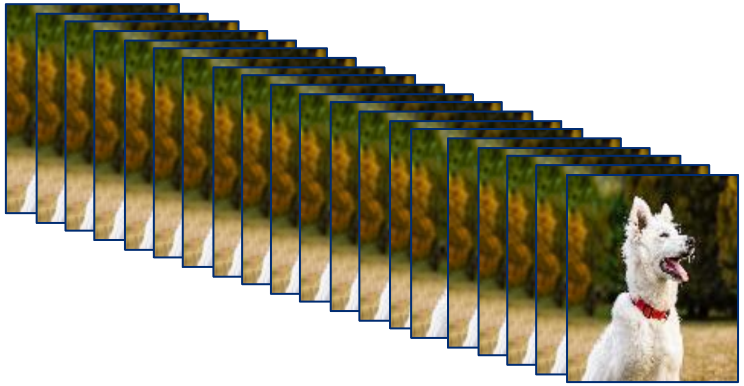


Limitations of self-attention

1. Computationally expensive
   - While they are more parallelizable than RNNs transformers tend to take more total compute resources
2. Memory footprint
   - Larger models can have trouble fitting in working memory
   - Running out of working memory significantly slows down computation!
3. Required training data
   - Transformers tend to need much larger training sets to learn the complex interactions they are designed to capture







What is a context window: What can LLMs ‘remember’?

- A context window is the amount of information that an LLM can actively consider at one time when generating a response
- Think of it as **Working Memory**
  - More context = better reasoning, summation, conversations
- It is not <u>permanent knowledge</u>
  - Information may be dropped

Human	LLM
Long-term Memory	Pre-training data
Working Memory	Context Window
Looking something up	Retrieval (RAG)

Transformers – Summary

Strengths:

- Wide context
- Highly parallelizable
- Broadly applicable

Limitations:

- Computationally expensive
- Memory footprint
- Requires training data